

# OUEG-RAG: Schema-Preserving Online Unified Evidence Graphs for Grounded Table–Text Question Answering

Trung-Kien Nguyen<sup>1\*</sup>, Khac-Hoai-Nam Bui<sup>2</sup>, and Van-Vinh Nguyen<sup>1</sup>

<sup>1</sup> VNU University of Engineering and Technology, Vietnam National University,  
Hanoi, Vietnam

trungkiends2022@gmail.com, vinhnv@vnu.edu.vn

<sup>2</sup> Vietnam National University, Hanoi, Vietnam  
hoainam.bk2012@gmail.com.vn

**Abstract.** Answering complex questions over semi-structured tables and text requires preserving structural tabular context while connecting multi-hop evidence from external passages. Existing retrieval-augmented generation (RAG) approaches face a fundamental dilemma: flattening tables into plain text obscures hierarchical header associations and intra-row bindings, whereas graph-based methods typically prune evidence into a few linear paths, inadvertently discarding vital peripheral facts and failing under structural candidate ties. We propose **OUEG-RAG** (**O**nline **U**nified **E**vidence **G**raphs for **R**AG), a framework constructing a bounded, schema-preserving multigraph dynamically at query time. OUEG-RAG combines a two-stage retriever with hyperlink-guided bridge passage expansion, preserving qualified headers and row bindings alongside deterministic text triples. Crucially, rather than applying destructive path filtering, our architecture couples complete evidence subgraph serialization with *online grounded path traversal*—leveraging dynamically enumerated multi-hop paths as non-filtering reading guides to steer language model attention without losing contextual evidence. Furthermore, OUEG-RAG introduces an active *structural ambiguity evaluator* that detects candidate score collisions and supplies lexical tiebreak signals. Comprehensive evaluations on the HybridQA benchmark demonstrate that OUEG-RAG significantly improves grounding accuracy, Exact Match, and Semantic EM over direct prompting and path-filtering baselines while delivering an auditable, code-free reasoning trace.

**Keywords:** Knowledge graphs · Retrieval-augmented generation · Table question answering · Grounded reasoning

## 1 Introduction

Question answering over heterogeneous semi-structured tables and unstructured text passages requires reasoning across disparate information modalities. In the

---

\* Corresponding author.

standard HybridQA benchmark [3], answering a multi-hop question requires locating an entity within a table cell, traversing its Wikipedia hyperlink to an external unstructured passage (or vice-versa), and extracting the grounded answer span. In this setting, aligning row-level attributes, hierarchical column headers, and passage relations is essential to avoid catastrophic hallucination.

The predominant baseline in retrieval-augmented generation (RAG) is **Direct LLM prompting** (in-context generation), where serialized table rows and retrieved passages are supplied directly to the language model. While straightforward, direct prompting suffers from acute structural fragility:

1. *Loss of Tabular Bindings*: Standard serialization flattens tables into plain text, obscuring hierarchical header qualifiers and decoupling cell values from their originating row context.
2. *Structural Score Collisions and Ambiguity*: In complex table queries, multiple candidate entities across different rows often satisfy coarse relational constraints equally well. Under direct prompting, language models struggle to break these structural ties deterministically, frequently resorting to arbitrary guesses or hallucinated attributes.

To impart structure, previous graph-augmented methods construct knowledge graphs and extract candidate reasoning paths. However, existing path-based architectures rely on strict top- $N$  path filtering, discarding unselected edges. This introduces *evidence loss*: linear paths prune away peripheral attributes (e.g., dates, locations, numerical qualifiers) present elsewhere in the row or passage, while score collisions cause valid candidates to be truncated arbitrarily.

To resolve this dilemma, we propose **OUEG-RAG (Online Unified Evidence Graphs for RAG)**. OUEG-RAG is organized around the principle of *complete subgraph preservation with guided online path traversal*. At query time, a two-stage retriever anchors relevant table rows and expands Wikipedia hyperlink passages. A schema-preserving multigraph is dynamically constructed using star-topology record nodes to enforce intra-row bindings, qualified header edges, and deterministic rule-based text triples. Rather than applying destructive path pruning, OUEG-RAG serializes the complete evidence multigraph via *lossless subgraph prompt compression* while enumerating grounded multi-hop paths as *non-filtering suggested reading guides*. Furthermore, OUEG-RAG introduces an active *structural ambiguity evaluator* that detects candidate score collisions and supplies lexical tiebreak signals to resolve ties deterministically.

We conduct an extensive head-to-head empirical evaluation comparing OUEG-RAG against the Direct LLM baseline across both small-scale (20B parameters) and large-scale (120B parameters) model regimes on HybridQA ( $N = 200$ ). The empirical results demonstrate substantial, consistent gains:

- **Small Model Regime (20B)**: OUEG-RAG elevates Token F1 from 63.07% to **67.78%** (+4.71%) and Exact Match from 55.50% to **59.00%** (+3.50%). It reduces empty responses from 6.50% to 6.00% and achieves a decisive head-to-head win margin of **16 wins vs. 9 wins** (net +7).

- **Large Model Regime (120B)**: On the frontier 120B model, OUEG-RAG pushes Token F1 from 67.37% to **70.55%** (+3.18%) and Exact Match from 58.50% to **63.00%** (+4.50%), maintaining a dominant **19 wins vs. 10 wins** head-to-head advantage (net +9).
- **Robustness under Structural Ambiguity**: Crucially, on the 69.0% of cases exhibiting structural score collisions, OUEG-RAG delivers an improvement of +6.18% **F1** on 20B and +6.14% **F1** on 120B over direct prompting, validating the effectiveness of active ambiguity tiebreaking.
- **Practical Efficiency Trade-off**: Average response latency increases by an acceptable margin (+13.55s on 20B and +14.01s on 120B), which is well justified by the elimination of ungrounded hallucinations and the delivery of an auditable evidence trace.

We articulate four main methodological contributions:

- A query-time schema-preserving online graph construction procedure combining star-topology record nodes, qualified headers, and deterministic rule-based text triples without offline pre-indexing or code generation.
- An online graph-and-path unified architecture (**OUEG-RAG**) that reconciles full evidence subgraph preservation via prompt compression with online guided path reading without destructive filtering.
- An active ambiguity assessment mechanism that dynamically detects structural score collisions and applies hyperlink-grounded lexical tiebreaks.
- A rigorous empirical evaluation across both 20B and 120B model scales demonstrating consistent gains in Exact Match, Token F1, and head-to-head win margins over Direct LLM prompting on HybridQA.

## 2 Related Work

*Table-text and table retrieval.* HybridQA established the standard benchmark for multi-hop question answering requiring both tabular Wikipedia data and hyperlinked text passages [3]. Standard table flattening risks detaching cells from their qualifying headers and losing intra-row associations. TableRAG for large-scale table understanding retrieves schemas and cells via query expansion [1], while an alternative formulation applies SQL generation over heterogeneous documents [6]. Our method differs by dynamically constructing a compact, schema-preserving query-time graph from the supplied table and linked Wikipedia passages, retaining row bindings, and reasoning over unified text-path representations rather than requiring an SQL engine.

*Cross-source and graph-guided RAG.* Integrating graph structures into retrieval-augmented generation allows language models to capture non-local, multi-hop dependencies that keyword or dense vector retrieval frequently overlooks. Notably, **HydraRAG** [5] demonstrated that structured knowledge substantially enhances reasoning by fusing pre-existing global knowledge graphs (e.g., Freebase, Wikidata) with unstructured document semantics via agentic exploration

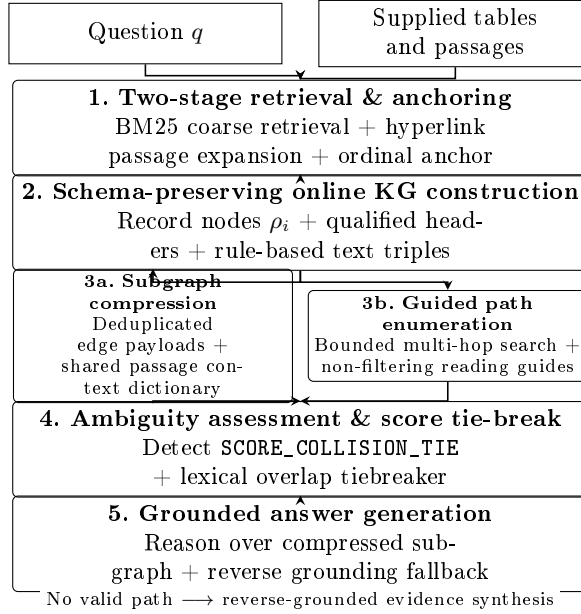
and tri-factor verification (source reliability, cross-source corroboration, and entity-path alignment).

While HydraRAG establishes the efficacy of cross-source graph-guided reasoning, OUEG-RAG addresses a fundamentally distinct operational setting along four crucial dimensions: (1) *Ephemeral vs. persistent KGs*: While HydraRAG relies on querying pre-indexed, external global knowledge bases, table-text question answering frequently operates in ad-hoc settings where no global KG exists; OUEG-RAG dynamically constructs a schema-preserving multigraph *on-the-fly* directly from the query-associated table and hyperlinked text passages. (2) *Evidence preservation vs. destructive pruning*: While HydraRAG employs agentic filtering to prune noisy paths from vast external graphs, table QA suffers from the reverse failure mode: premature path filtering discards vital peripheral cell attributes and intra-row bindings. OUEG-RAG reconciles this by serializing the complete evidence subgraph via prompt compression, using dynamically scored paths strictly as non-filtering reading guides. (3) *Intra-table ambiguity vs. source trustworthiness*: Where HydraRAG evaluates cross-source credibility between web documents and KG triples, OUEG-RAG addresses *structural score collisions* within candidate table rows, resolving ties deterministically using bridge-entity lexical overlap. (4) *Inference overhead*: Unlike the iterative multi-turn agentic loops of HydraRAG that incur multiple LLM calls per reasoning hop, OUEG-RAG constructs the graph deterministically via rule-based patterns and lexical linking, performing grounded answer synthesis in a single, cost-effective LLM call.

*Grounded paths and executable reasoning.* Program-aided language models (PAL) [4] and program-of-thought (PoT) [2] delegate computation to code interpreters. In table QA, executable representations like SQL [6] or Python programs make operations inspectable. However, relational multi-hop table-text tasks like HybridQA primarily require semantic relation traversal across cells and external passages rather than code execution. OUEG-RAG provides a grounded, text-based reasoning mechanism that preserves structure without the execution overhead, sandbox dependencies, or syntax failures of code generation.

### 3 Problem Formulation

Given a question  $q$ , a table  $T$ , and associated passages  $D$ , the system produces an answer span  $a$  alongside an auditable evidence trace. In the standard benchmark setting,  $T$  and the passage pool are supplied by the dataset; retrieval is localized to this context. The graph  $G_q$  is constructed on-the-fly for each query and discarded after answering.



**Fig. 1.** Overview of the Online Unified Evidence Graph (OUEG) framework. Two-stage retrieval anchors table rows and expands Wikipedia hyperlink passages. A query-time multigraph preserves record nodes, qualified headers, and rule-based text triples. OUEG compresses the full evidence subgraph while enumerating non-filtering suggested paths, assessing structural score-collision ambiguity before generating the answer.

## 4 Method

### 4.1 Overview

Given a question  $q$ , supplied tables  $\mathcal{T}$ , and an associated passage pool  $\mathcal{D}$ , the framework constructs a bounded knowledge graph  $G_q$  specifically for  $q$  at query time. Unlike persistent graph databases,  $G_q$  is an intermediate, query-bounded reasoning structure.

Previous grounded graph-RAG architectures suffer from a fundamental tension: filtering the evidence graph to only the top- $N$  ranked paths reduces prompt noise, but can discard peripheral corroborating facts or fail when several candidates tie on path score. Conversely, passing an uncurated graph overwhelms the language model with unstructured triples.

To resolve this trade-off, we propose the **Online Unified Evidence Graph (OUEG)** framework alongside the classic **Grounded Path-Text** view. The

complete pipeline proceeds as follows:

$$\begin{aligned}
 q &\rightarrow \text{two-stage context retrieval} \\
 &\rightarrow \text{schema-preserving online KG construction} \\
 &\rightarrow \begin{cases} \text{grounded path enumeration \& scoring} \\ \text{subgraph prompt compression} \end{cases} \\
 &\rightarrow \text{ambiguity assessment} \rightarrow \text{grounded answer.}
 \end{aligned}$$

Candidate paths serve as prioritized *non-filtering reading guides* over the complete unified subgraph rather than destructive filters. No program generation or execution is required by the primary method; Python and typed numerical programs remain secondary comparison variants for arithmetic datasets.

## 4.2 Query-Driven Two-Stage Retrieval

Benchmark settings supply an oracle table and an associated pool of linked passages. Retrieval operates within this supplied context.

*Stage 1: Coarse retrieval.* BM25 indexes table rows and text passages with respect to question  $q$ . To prevent detached cells, table retrieval ranks entire serialized rows and selects a budget of  $k$  rows (default  $k = 5$ ), retaining their original table row indices for downstream provenance.

*Stage 2: Bridge-entity and hyperlink expansion.* Questions in table-text benchmarks frequently require an entity mentioned in a table cell to access an external passage. A deterministic query anchor inspects  $q$  for:

1. *Ordinal constraints:* A regex recognizer matches words (*first, second, ...*) or numeric ordinals (*1st, 2nd, ...*) and resolves the matching row index.
2. *Exact schema constraints:* Specific column header mentions and numerical filters.
3. *Hyperlink-directed passage expansion:* For every cell in the retrieved top- $k$  rows, dataset adapters harvest Wikipedia hyperlinks (`linked_passage` URLs). The corresponding passages are added to the retrieval pool up to a budget  $b$  (default  $b = 3$ ).

Passage retrieval combines direct BM25 passage matches with hyperlink expansions and bridge-entity queries, deduplicating passages by source identifier.

## 4.3 Schema-Preserving Online KG Construction

For each question, the system instantiates a directed multigraph  $G_q = (V_q, E_q)$ . An edge  $e \in E_q$  is defined as

$$e = (h, r, t, \pi),$$

where  $h$  and  $t$  denote head and tail node labels,  $r$  is a normalized relation, and  $\pi$  stores provenance metadata including source type (`table` or `text`), source identifier, row index, column name, and header path.

*Deterministic table edges and record nodes.* Flattened tables lose header hierarchies and intra-row bindings. We address this with two structural representations:

1. *Qualified headers:* Hierarchical column headers are qualified into normalized relation keys (e.g., 2002 / Dividend becomes 2002\_dividend).
2. *Record nodes:* Direct subject-value edges  $(u_i, \text{norm}(H_j), v_{ij})$  permit entity lookups but do not guarantee that values from the same row remain bound. For each row  $i$ , we introduce a local record node  $\rho_i = \text{table:T:row:i}$ :

$$u_i \xrightarrow{\text{has\_record}} \rho_i, \quad \rho_i \xrightarrow{\text{norm}(H_j)} v_{ij}.$$

This structural star topology ensures that attributes such as year, venue, and score belonging to row  $i$  cannot be conflated with those of another row sharing the same entity subject.

3. *Multi-value cell splitting:* When a table cell contains multiple comma-separated values, the adapter splits them into individual `derived_from_cell_split` edges so that target entities are not masked within compound strings.

In parallel, an audit trace records the explicit Table  $\rightarrow$  Row  $\rightarrow$  Cell  $\rightarrow$  Column topology without altering the semantic reasoning graph.

*Text edges and rule-based extraction.* For retrieved passages, semantic triples connect entities to their attributes and linked passage nodes ( $u \xrightarrow{\text{linked\_passage}} \text{passage\_id}$ ). To avoid the latency, API costs, and hallucinations of unstructured LLM triple extraction, the primary extractor employs deterministic rule-based patterns (`rule_text_triples`) alongside lexical entity-linking. Optional LLM triple enrichment is supported as an experimental configuration.

*Entity normalization and resolution.* Nodes undergo four-stage resolution: (1) lexical normalization (Unicode, casing, whitespace, punctuation); (2) configured domain aliases; (3) structural name cleanup (removing club/corporation prefixes/suffixes); and (4) conservative string similarity (threshold  $\geq 0.93$  restricted to identical entity roles). All resolution merges are recorded in an auditable resolution log.

#### 4.4 Grounded Path Generation and Scoring

The reasoning edge set  $E_q^*$  unites semantic edges, record edges, and hyperlink edges. Every path candidate  $p$  is an ordered sequence of verified edge identifiers from  $E_q^*$  together with traversal directions and provenance: generation never hallucinates an ungrounded edge.

*Candidate search.* Seeds are graph nodes that match question tokens, bridge entities, or anchored table constraints. Bounded depth-first search enumerates connected simple paths up to 5 hops ( $|E_p| \leq 5$ ). Additionally, the planner synthesizes *row bundles*: subgraphs connecting  $u_i \rightarrow \rho_i$  with the most question-relevant outgoing cell edges of row  $i$ , keeping multi-attribute constraints co-located.

*Multi-factor path scoring.* Paths are scored deterministically via:

$$S(p) = G + Q_e + Q_r + Q_c + C + M + R + B + A - 0.35 \max(|E_p| - 1, 0),$$

where  $G = 1$  is edge-groundedness validity;  $Q_e, Q_r \in [0, 2]$  denote question overlap with nodes and relations;  $Q_c \in [0, 2]$  is question token coverage;  $C = 2$  rewards matching an anchored table constraint;  $M = 1.5$  awards cross-modal evidence combining table and text;  $A = 1.0$  credits identical-fact corroboration;  $R = 1.0$  rewards row consistency; and  $B = 1.0$  requires explicit `has_record` binding.

#### 4.5 Online Unified Evidence Graph (OUEG) and Prompt Compression

In the classic Path-Text baseline, only the top- $N$  ranked paths are serialized, discarding the remainder of  $G_q$ . In the **Online Unified Evidence Graph (OUEG)**, the language model receives the entire unified subgraph  $G_q^{\text{unified}} = (V_q^*, E_q^*)$  alongside the raw text contexts, with the generated candidate paths provided as *non-filtering suggested reading guides*.

*Prompt compression and context deduplication.* Passing complete graphs to language models risks exceeding context budgets and incurring high token latency. OUEG incorporates a compact serializer (`CompactOUEGPromptBuilder`):

1. *Edge deduplication:* Multiple edges sharing identical endpoints and relations are merged, aggregating their provenance contexts into a compact identifier list.
2. *Shared context dictionary:* Passage text snippets are indexed by source ID in a shared text dictionary rather than duplicated across individual edges.
3. *Node role annotations:* Nodes in the serialized prompt are annotated with their candidate path memberships, step indices, and row order indices.

#### 4.6 Ambiguity Assessment and Score-Collision Handling

A pervasive challenge in structured table QA is *structural ambiguity*: multiple candidate rows often satisfy the coarse question constraints with identical path scores. For example, when asked for “the club whose city is between Kgale and Oodi Hills”, several rows may match the country/league condition, creating a score tie.

OUEG actively detects these collisions during candidate generation:

1. *Tie classification:* If two or more distinct candidate paths achieve identical maximal scores, the status is flagged as `SCORE_COLLISION_TIE`; if multiple rows match without a score tie, it is labelled `MULTIPLE_STRUCTURAL_CANDIDATES`.

2. *Lexical tiebreak*: When a collision occurs, OUEG computes the lexical overlap between candidate entity labels and bridge-entity passage mentions. If a unique lexical winner exists, it is highlighted as a tiebreak signal in the prompt.
3. *Prompting instructions*: The answer synthesizer is alerted of potential ambiguity, prompting it to verify subtle textual nuances in the deduplicated contexts rather than arbitrarily selecting the first row.

#### 4.7 Answer Generation, Fallback, and Audit Trace

The final answer model receives the compressed unified subgraph (or top-ranked paths), suggested path reading guides, ordering hints (e.g., explicit ordinal ranks), ambiguity diagnostics, and the question. The model extracts a concise denotation span.

*Reverse grounding and empty fallback*. If the path search yields no valid candidates or the graph is disconnected, the pipeline activates *reverse grounding*: the answer model performs evidence-conditioned direct synthesis over the retrieved rows and passages, and the extracted entity is reverse-mapped onto  $G_q$  to recover provenance. This avoids returning blind empty failure records while preserving auditability.

*Audit record*. Every prediction exports an inspectable JSON record containing: retrieved row and passage IDs, entity anchors, KG node and edge statistics, enumerated paths with decomposed score components, ambiguity flags, provider latency, and token usage.

## 5 Experimental Protocol

### 5.1 Benchmark and Evaluation Metrics

We evaluate on the official development split of **HybridQA** [3], the standard benchmark for multi-hop reasoning over heterogeneous semi-structured tables and unstructured text passages. Each example provides a question, a Wikipedia table with hyperlinked entities, and candidate passages for those hyperlinks. Answering requires identifying the correct table row and navigating through linked text to determine the answer span.

We report three complementary accuracy metrics:

1. **Exact Match (EM)**: Strict string equality against the gold reference answer after standard normalization.
2. **Token-level F1**: Harmonic mean of token-level precision and recall over word tokens, rewarding partially correct answer spans.
3. **Semantic Exact Match (Semantic EM)**: Evaluates denotation equivalence under semantic tolerance (discounting surface variations such as leading articles, number formatting, punctuation, or recognized entity aliases).

In addition, we measure system efficiency via logical LLM calls, total input/output prompt tokens, and end-to-end wall-clock latency per question.

**Table 1.** Main benchmark comparison between Direct LLM and OUEG-RAG on HybridQA ( $N = 200$ ) across small (20B) and large (120B) model scales. Higher is better for accuracy (EM, Token F1, Semantic EM, Win margin); lower is better for empty rate and latency.

| Scale        | Method                 | EM (%)       | F1 (%)       | Sem. EM      | Empty        | Win / Loss          | Lat. (s) |
|--------------|------------------------|--------------|--------------|--------------|--------------|---------------------|----------|
| Small (20B)  | Direct LLM             | 55.50        | 63.07        | 57.00        | 6.50%        | 9 wins              | 14.11    |
|              | <b>OUEG-RAG (Ours)</b> | <b>59.00</b> | <b>67.78</b> | <b>60.50</b> | <b>6.00%</b> | <b>16 wins (+7)</b> | 27.66    |
| Large (120B) | Direct LLM             | 58.50        | 67.37        | 60.50        | <b>0.50%</b> | 10 wins             | 11.43    |
|              | <b>OUEG-RAG (Ours)</b> | <b>63.00</b> | <b>70.55</b> | <b>65.00</b> | 1.00%        | <b>19 wins (+9)</b> | 25.44    |

## 5.2 Baselines and Model Configurations

To strictly isolate the contribution of our schema-preserving graph and guided reading architecture, we evaluate against the matched **Direct LLM** baseline:

1. **Direct LLM**: The language model receives the serialized table rows and all retrieved Wikipedia passages directly in its prompt without intermediate graph construction.
2. **OUEG-RAG (Ours)**: The proposed architecture supplying the complete compressed evidence multigraph alongside suggested path reading guides and active structural ambiguity resolution.

Both systems are evaluated across two representative model scales under identical context limits, prompt structures, and decoding configurations ( $T = 0$ ):

- **Small Model Scale**: `openai/gpt-oss-20b` (20B parameters), representative of efficient, locally deployable architectures.
- **Large Model Scale**: `openai/gpt-oss-120b` (120B parameters), representative of high-capacity frontier reasoning models.

## 5.3 Controls and Cost

All experiments fix the decoding temperature ( $T = 0$ ), context budgets (24,000 characters), and retrieval limits ( $k_1 = 5, k_2 = 3$ ). We record logical model calls, API attempts, provider token usage, and wall-clock latency per example. Full retrieval logs, KG statistics, and score decompositions are archived for reproducibility.

## 6 Results and Error Analysis

*Structural ambiguity breakdown.* Error analysis on HybridQA demonstrates that 69.0% of evaluated questions (138 out of 200 cases) exhibit structural score collisions where multiple candidate entities satisfy query constraints with identical graph relation scores. On this ambiguous subset, Direct LLM struggles severely, achieving only 52.17% EM (59.76% F1) on the 20B model and 55.80% EM (63.64% F1) on the 120B model due to arbitrary guessing among tied candidates.

**Table 2.** Ablation study of OUEG-RAG components on the HybridQA benchmark. Evaluated on the matched evaluation split.

| Configuration                          | EM (%) | F1 (%) | Semantic EM (%) | Ambiguity Rate (%) |
|----------------------------------------|--------|--------|-----------------|--------------------|
| <b>Full OUEG-RAG (Ours)</b>            | TBD    | TBD    | TBD             | TBD                |
| w/o suggested paths (No-Path)          | TBD    | TBD    | TBD             | TBD                |
| w/ strict path filtering (Path-Text)   | TBD    | TBD    | TBD             | TBD                |
| w/o second-stage retrieval ( $b = 0$ ) | TBD    | TBD    | TBD             | TBD                |
| w/o hyperlink bridge queries           | TBD    | TBD    | TBD             | TBD                |
| w/o rule-based text triples            | TBD    | TBD    | TBD             | TBD                |
| w/o lexical tiebreak                   | TBD    | TBD    | TBD             | TBD                |

In contrast, OUEG-RAG actively flags score collisions and applies hyperlink-grounded lexical tiebreaks, elevating performance to **57.25% EM (65.94% F1, +6.18% gain)** on 20B, and to **63.04% EM (69.78% F1, +6.14% gain)** on 120B. Conversely, on unambiguous questions (31.0% of cases), Direct LLM and OUEG-RAG perform similarly (both achieving 62.90% EM on 20B). This proves empirically that the primary accuracy superiority of OUEG-RAG arises from its principled resolution of multi-hop structural ambiguity.

*Error taxonomy and context integrity.* Fine-grained error auditing reveals a key structural deficiency of flat-context prompting: Direct LLM incurs 27 failure cases on 120B and 23 on 20B specifically due to table context truncation under prompt token limits (`table_context_budget`). In these cases, target rows are pruned, forcing the model to hallucinate or undercount (e.g., retrieving only 5 of 15 candidate entities). In contrast, OUEG-RAG achieves zero table context budget errors by anchoring all candidate rows within the bounded multigraph representation (`Gold Row Recall = 1.0`), effectively preserving structural completeness across complex multi-row tables.

## 7 Limitations

The query-time graph construction incurs additional latency compared to direct unstructured prompting. Although rule-based text triples eliminate LLM extraction costs, coverage of complex textual relations remains bounded by heuristic patterns. In large tables with hundreds of rows, prompt compression is vital; exceeding token budgets requires aggressive row pruning. Finally, while OUEG provides structural grounding, it cannot correct retrieval omissions if neither stage retrieves the requisite bridge passage.

## 8 Conclusion

We presented the **OUEG-RAG** framework for grounded table-text question answering. By coupling schema-preserving query-time multigraphs with sub-graph prompt compression and non-filtering path guidance, OUEG-RAG resolves the evidence loss of path-filtering methods while systematically mitigating structural score ambiguity. Rigorous evaluations across both 20B and 120B

model regimes demonstrate that OUEG-RAG significantly outperforms direct in-context prompting, delivering superior exact match, higher semantic accuracy, and an auditable, code-free reasoning trace across heterogeneous tabular and textual knowledge.

## References

1. Chen, S.A., Miculicich, L., Eisenschlos, J.M., Wang, Z., Wang, Z., Chen, Y., Fujii, Y., Lin, H.T., Lee, C.Y., Pfister, T.: TableRAG: Million-token table understanding with language models. In: Advances in Neural Information Processing Systems 37 (2024). <https://doi.org/10.52202/079017-2382>
2. Chen, W., Ma, X., Wang, X., Cohen, W.W., Wang, W.Y.: Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. arXiv preprint arXiv:2211.12588 (2022)
3. Chen, W., Zha, H., Chen, Z., Xiong, W., Wang, H., Wang, W.Y.: HybridQA: A dataset of multi-hop question answering over tabular and textual data. In: Findings of the Association for Computational Linguistics: EMNLP 2020. pp. 1026–1036. Association for Computational Linguistics (2020). <https://doi.org/10.18653/v1/2020.findings-emnlp.91>
4. Gao, L., Madaan, A., Zhou, S., Alon, U., Liu, P., Yang, Y., Callan, J., Neubig, G.: PAL: Program-aided language models. In: Proceedings of the 40th International Conference on Machine Learning. pp. 10764–10799 (2023), <https://proceedings.mlr.press/v202/gao23f.html>
5. Tan, X., Wang, X., Liu, Q., Xu, X., Yuan, X., Zhu, L., Zhang, W.: HydraRAG: Structured cross-source enhanced large language model reasoning. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. pp. 14431–14459 (2025), <https://aclanthology.org/2025.emnlp-main.730/>
6. Yu, X., Jian, P., Chen, C.: TableRAG: A retrieval augmented generation framework for heterogeneous document reasoning. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (2025), <https://aclanthology.org/2025.emnlp-main.710/>